Vincent Onoja

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PROFESSIONAL SUMMARY

Experimental researcher and Ph.D. candidate specializing in the design, construction, and operation of precision laser diagnostic systems. Experienced with optics, high-speed imaging, custom test hardware, pressure systems, and automated data acquisition, with a proven track record of validating complex physical systems through comparison with theoretical and computational predictions. Secured $1.475M in competitive federal research funding.

TECHNICAL SKILLS

- **Experimental Systems & Laser Diagnostics:** Pulsed laser operation and sheet-forming optics; optical alignment; laser–camera timing and synchronization; camera calibration; high-speed imaging (10 kHz); planar and stereoscopic PIV; wind tunnel testing; pressure/temperature instrumentation (thermocouples, transducers)
- **Instrumentation, DAQ & Automation:** LabVIEW (multi-channel synchronized DAQ), Python-based test automation, signal conditioning, experiment control and troubleshooting
- **Hardware Design & Fabrication:** SolidWorks, CATIA V6/3DEXPERIENCE, GD&T, ASME pressure vessel design, 3D printing (FDM/SLA), design of optical-access test hardware
- **Scientific Computing & ML:** Python (NumPy, SciPy, Pandas), TensorFlow/Keras, PyTorch, CNNs, Scientific Machine Learning (SciML), MATLAB, Git/GitHub
- **Physics-Based Simulation:** ANSYS Fluent, STAR-CCM+, turbulence modeling (RANS/LES), thermodynamics, simulation–experiment correlation

PROFESSIONAL EXPERIENCE

University of Cincinnati | Advanced Laboratory for Fluids and Acoustics (ALFA)

Graduate Research Assistant ***Aug. 2020 – Present***

- Designed, built, and operate a flow mixing rig with custom nozzle test sections engineered for optical access (multi-window design, full GD&T drawings), enabling laser-based measurements inside pressurized hardware.
- Operate and align a 10 kHz high-speed PIV system end-to-end: pulsed laser and sheet-forming optics, laser–camera timing/synchronization, calibration, seeding, and troubleshooting of signal-quality and alignment issues.
- Developed automated data acquisition systems (LabVIEW, Python) synchronizing temperature, pressure, and velocity-field measurements across multiple channels for repeatable, high-throughput experimental campaigns.
- • Led a multi-fidelity validation campaign using experimental ground truth (PIV and pitot measurements) to benchmark computational predictions from industry RANS simulations (ANSYS Fluent), high-fidelity LES (U.S. Navy WMLES), and developmental research solvers (MIT's SANS code).
- Secured $1.475M in competitive federal funding (ONR/AFOSR) as lead technical contributor, defining the roadmap for experimentally verifying next-generation naval propulsion simulation tools.
- Coordinated fabrication with machine-shop staff and mentored junior lab members on PIV operation, instrumentation, and DAQ.

Tesla, Inc.

Hardware Test Engineering Intern ***Jan. 2023 – Apr. 2023***

- Developed Python automation for seat durability testing, reducing test cycle time in a fast-paced hardware development environment.
- Designed a vibration test fixture for the Cybertruck program using CATIA V6/3DEXPERIENCE.
- Led an organization-wide root-cause investigation into a persistent seat-yellowing defect; documented findings and presented to key Tesla executives.

LEADERSHIP & PROFESSIONAL SERVICE

- AIAA Inlets, Nozzles, and Propulsion Systems Integration (INPSI) Technical Committee **(May 2025 – Present)**: one of ~50 national experts on the governing body defining technical standards for the aerospace propulsion integration sector.
- Peer Reviewer, AIAA Journal of Aircraft (2024–2025); reviewer for multiple AIAA AVIATION and SCITECH abstracts.
- Author of 1 peer-reviewed journal publication, 1 journal article under review (Aerospace Science and Technology), and 4 conference papers (AIAA SciTech, AIAA AVIATION, ISPIV — Japan) in experimental fluid dynamics and CFD validation (see google scholar)

EDUCATION

Ph.D. in Aerospace Engineering | University of Cincinnati, OH | Defense: **Aug., 26 2026**

B.S. in Mechanical Engineering | University of Ibadan, Nigeria | Dec.**2015** | First Class Honors